

Trajectory Uncertainty Calibration for Multimodal VLMs

Complete Benchmark & Transfer LaTeX Table Compendium

M3-LLaVA (7B) & MQT-LLaVA (7B) Evaluations

September 14, 2026

Abstract

This document compiles all 25 publication-ready LaTeX tables across Groups 1–7 for local inspection before Overleaf integration. All tables follow the standardized 4-metric schema: ECE (%) ↓, Ada-ECE (%) ↓, AUROC ↑, and Brier score ↓, featuring Option A ranking (**bold** Rank 1, *italic* Rank 2) and Trajectory Platt (5D) evaluation.

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Table 1: **Zero-Shot Cross-Domain Calibration Transfer (M3-LLaVA 7B)**. Pairwise zero-shot calibration transfer from source datasets across held-out target vision-language benchmarks without target domain supervision. Evaluated across Expected Calibration Error (**ECE** ↓), Adaptive ECE (**Ada-ECE** ↓), AUROC (↑), and Brier score (↓). **Bold**: Rank 1; *italic*: Rank 2 among transfer methods.

Source: VQAv2 (General Scene VQA)												
Method	DocVQA (Document Text)				ChartQA (Diagrams)				TextVQA (Scene Text)			
	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓
IN-DOMAIN CALIBRATION (PLATT 5D)	0.0231	0.0239	0.732	0.0409	0.0343	0.0388	0.826	0.0681	0.0435	0.0591	0.839	0.1604
Uncalibrated (NC)	0.7421	0.7421	0.777	0.6371	0.7168	0.7168	0.807	0.6208	<i>0.0765</i>	<i>0.0770</i>	0.840	0.1689
Temperature Scaling (TS)	<i>0.6865</i>	<i>0.6865</i>	0.777	<i>0.5389</i>	<i>0.6622</i>	<i>0.6622</i>	0.807	<i>0.5293</i>	0.1123	0.1136	0.840	<i>0.1741</i>
Platt Scaling (1D)	0.7496	0.7496	0.777	0.6108	0.7199	0.7199	0.807	0.5956	0.1788	0.1820	0.840	0.2074
Platt 5D (Ours)	0.5616	0.5616	<i>0.775</i>	0.3982	0.5676	0.5676	<i>0.765</i>	0.4251	0.0652	0.0650	<i>0.803</i>	0.1807
Source: VizWiz-VQA (Assistive / Blurry)												
Method	ChartQA (Diagrams)				DocVQA (Document Text)				TextVQA (Scene Text)			
	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓
IN-DOMAIN CALIBRATION (PLATT 5D)	0.0343	0.0388	0.826	0.0681	0.0231	0.0239	0.732	0.0409	0.0435	0.0591	0.839	0.1604
Uncalibrated (NC)	0.7168	0.7168	0.807	0.6208	0.7421	0.7421	<i>0.777</i>	0.6371	<i>0.0765</i>	<i>0.0770</i>	0.840	0.1689
Temperature Scaling (TS)	0.6916	0.6916	0.807	0.5768	0.7167	0.7167	<i>0.777</i>	0.5902	0.0832	0.0815	0.840	0.1693
Platt Scaling (1D)	<i>0.6148</i>	<i>0.6148</i>	0.807	<i>0.5094</i>	<i>0.6367</i>	<i>0.6367</i>	<i>0.777</i>	<i>0.5203</i>	0.0829	0.0885	0.840	<i>0.1675</i>
Platt 5D (Ours)	0.6114	0.6114	<i>0.806</i>	0.5084	0.6308	0.6308	0.779	0.5176	0.0762	0.0762	<i>0.837</i>	0.1671
Source: TextVQA (Scene Text)												
Method	ChartQA (Diagrams)				DocVQA (Document Text)				VizWiz-VQA (Assistive)			
	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓
IN-DOMAIN CALIBRATION (PLATT 5D)	0.0343	0.0388	0.826	0.0681	0.0231	0.0239	0.732	0.0409	0.0627	0.0675	0.837	0.1636
Uncalibrated (NC)	0.7168	0.7168	<i>0.807</i>	0.6208	0.7421	0.7421	<i>0.777</i>	0.6371	0.1656	0.1656	0.792	0.2099
Temperature Scaling (TS)	0.7097	0.7097	<i>0.807</i>	0.6080	0.7350	0.7350	<i>0.777</i>	0.6236	0.1622	0.1622	0.792	0.2086
Platt Scaling (1D)	<i>0.6733</i>	<i>0.6733</i>	<i>0.807</i>	<i>0.5741</i>	<i>0.6972</i>	<i>0.6972</i>	<i>0.777</i>	<i>0.5887</i>	0.0883	0.0865	0.792	0.1875
Platt 5D (Ours)	0.6653	0.6653	0.811	0.5631	0.6756	0.6756	0.784	0.5629	<i>0.1179</i>	<i>0.1182</i>	<i>0.784</i>	<i>0.1980</i>

1 Main Manuscript Tables (Groups 5, 2, 6, 7, 4)

1.1 Table 2: Zero-Shot Cross-Domain Calibration Transfer (M3-LLaVA 7B)

Table 2: **Zero-Shot Cross-Domain Calibration Transfer (MQT-LLaVA 7B)**. Pairwise zero-shot calibration transfer from source datasets across held-out target vision-language benchmarks without target domain supervision. Evaluated across Expected Calibration Error (**ECE** ↓), Adaptive ECE (**Ada-ECE** ↓), AUROC (↑), and Brier score (↓). **Bold**: Rank 1; *italic*: Rank 2 among transfer methods.

Source: VQAv2 (General Scene VQA)												
Method	DocVQA (Document Text)				ChartQA (Diagrams)				TextVQA (Scene Text)			
	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓
IN-DOMAIN CALIBRATION (PLATT 5D)	0.0105	0.0386	0.585	0.0381	0.0236	0.0627	0.713	0.0920	0.0587	0.0783	0.714	0.1985
Uncalibrated (NC)	0.5219	0.5219	<i>0.616</i>	0.4003	0.6416	0.6416	<i>0.428</i>	0.5916	0.3157	0.3157	<i>0.718</i>	0.3118
Temperature Scaling (TS)	<i>0.5083</i>	<i>0.5083</i>	<i>0.616</i>	<i>0.3282</i>	<i>0.5725</i>	<i>0.5725</i>	<i>0.428</i>	<i>0.4706</i>	<i>0.2773</i>	<i>0.2773</i>	<i>0.718</i>	<i>0.2782</i>
Platt Scaling (1D)	0.6147	0.6147	<i>0.616</i>	0.4294	0.6362	0.6362	<i>0.428</i>	0.5236	0.3573	0.3573	<i>0.718</i>	0.3284
Platt 5D (Ours)	0.4418	0.4418	0.641	0.2676	0.5288	0.5280	0.573	0.4018	0.2292	0.2276	0.742	0.2485
Source: VizWiz-VQA (Assistive / Blurry)												
Method	ChartQA (Diagrams)				DocVQA (Document Text)				TextVQA (Scene Text)			
	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓
IN-DOMAIN CALIBRATION (PLATT 5D)	0.0236	0.0627	0.713	0.0920	0.0105	0.0386	0.585	0.0381	0.0587	0.0783	0.714	0.1985
Uncalibrated (NC)	0.6416	0.6416	<i>0.428</i>	0.5916	0.5219	0.5219	<i>0.616</i>	0.4003	0.3157	0.3157	<i>0.718</i>	0.3118
Temperature Scaling (TS)	0.5368	0.5368	<i>0.428</i>	0.4167	0.4989	0.4989	<i>0.616</i>	0.3051	0.2536	0.2536	<i>0.718</i>	0.2655
Platt Scaling (1D)	<i>0.3754</i>	<i>0.3779</i>	<i>0.428</i>	<i>0.3353</i>	<i>0.2343</i>	<i>0.2343</i>	<i>0.616</i>	<i>0.1495</i>	<i>0.1427</i>	<i>0.1403</i>	<i>0.718</i>	<i>0.2342</i>
Platt 5D (Ours)	0.3214	0.3250	0.448	0.2845	0.1899	0.1899	0.637	0.1181	0.1273	0.1273	0.736	0.2180
Source: TextVQA (Scene Text)												
Method	ChartQA (Diagrams)				DocVQA (Document Text)				VizWiz-VQA (Assistive)			
	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓	ECE ↓	Ada-ECE ↓	AUROC ↑	Brier ↓
IN-DOMAIN CALIBRATION (PLATT 5D)	0.0236	0.0627	0.713	0.0920	0.0105	0.0386	0.585	0.0381	0.0541	0.0598	0.803	0.1454
Uncalibrated (NC)	0.6416	0.6416	<i>0.428</i>	0.5916	0.5219	0.5219	<i>0.616</i>	0.4003	0.3166	0.3166	0.808	0.2486
Temperature Scaling (TS)	0.4301	0.4301	<i>0.428</i>	0.2857	0.4704	0.4704	<i>0.616</i>	0.2601	0.2776	0.2837	0.808	0.2434
Platt Scaling (1D)	<i>0.2724</i>	<i>0.2776</i>	<i>0.428</i>	<i>0.1983</i>	<i>0.2501</i>	<i>0.2501</i>	<i>0.616</i>	<i>0.1115</i>	<i>0.1046</i>	<i>0.1152</i>	0.808	<i>0.1555</i>
Platt 5D (Ours)	0.2359	0.2382	0.553	0.1738	0.1967	0.1967	0.645	0.0913	0.0855	0.0845	<i>0.761</i>	0.1540

1.2 Table 3: Zero-Shot Cross-Domain Calibration Transfer (MQT-LLaVA 7B)

Table 3: **Macro-Averaged Calibration Benchmark of 8 Methods Across All 14 Datasets (M3-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	Macro ECE (%) ↓	Macro Ada-ECE (%) ↓	Macro AUROC ↑	Macro Brier ↓
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	40.33	40.33	0.671	0.3950
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	21.93	22.68	0.671	0.2213
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	3.24	<i>5.70</i>	0.682	<i>0.1357</i>
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	<i>4.25</i>	5.49	0.717	0.1322
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	19.24	-	0.675	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	28.25	-	0.641	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	24.74	-	0.694	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	17.34	-	<i>0.698</i>	-

Table 4: **Macro-Averaged Calibration Benchmark of 8 Methods Across All 14 Datasets (MQT-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	Macro ECE (%) ↓	Macro Ada-ECE (%) ↓	Macro AUROC ↑	Macro Brier ↓
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	32.32	31.76	0.682	0.3055
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	23.45	24.13	0.682	0.2163
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	5.88	<i>7.82</i>	0.686	<i>0.1365</i>
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	<i>6.06</i>	7.18	0.748	0.1334
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	22.77	-	0.666	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	33.37	-	0.639	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	29.57	-	0.689	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	21.27	-	<i>0.691</i>	-

1.3 Group 2: Macro-Averaged Calibration Benchmark Across All 14 Datasets

Table 5: **Zero-Shot Cross-Domain Source Anchor Leaderboard.** Performance of Trajectory Platt (5D) across candidate source anchors vs. standard transfer baselines across 12 unseen target benchmarks. Evaluated zero-shot on M3-LLaVA (7B) and MQT-LLaVA (7B). **Bold:** Rank 1; *Italic:* Rank 2 among transfer configurations. ↓: lower is better; ↑: higher is better.

Configuration / Source Anchor	M3-LLaVA (7B)				MQT-LLaVA (7B)				Dual-Arch
	ECE (%) ↓	Ada-ECE (%) ↓	AUROC ↑	Brier ↓	ECE (%) ↓	Ada-ECE (%) ↓	AUROC ↑	Brier ↓	Total Wins ↑
IN-DOMAIN CALIBRATION (PLATT 5D)	4.25	5.49	0.717	0.1322	6.06	7.18	0.748	0.1334	—
<i>Zero-Shot Transfer Baselines (Macro Mean Across All Pairs)</i>									
Naive Confidence (NC)	40.84	40.85	0.676	0.4032	33.57	33.52	0.693	0.3167	—
Temperature Scaling (TS)	33.46	35.01	0.676	0.3107	29.64	30.00	0.693	0.2564	—
Platt Scaling (1D)	37.93	38.12	0.645	0.3639	30.80	30.84	0.596	0.2908	—
<i>Trajectory Platt (5D) Candidate Source Anchors</i>									
VQAv2	38.12	38.12	<i>0.677</i>	0.3564	32.15	31.99	0.676	0.2891	22/24
VizWiz-VQA	41.01	41.03	0.666	0.3886	20.11	20.09	<i>0.691</i>	<i>0.2056</i>	<i>19/24</i>
DocVQA	29.88	30.06	0.680	0.3313	36.81	36.82	0.666	0.3521	18/24
TextVQA	41.79	41.80	0.669	0.4046	<i>20.45</i>	<i>20.56</i>	0.671	0.1879	18/24
ChartQA	<i>31.54</i>	<i>31.53</i>	0.676	<i>0.3307</i>	34.49	34.34	0.519	0.3322	18/24
POPE	43.22	43.20	0.671	0.4201	33.16	33.07	0.678	0.3133	10/24

1.4 Group 6: Zero-Shot Source Anchor Leaderboard

Table 6: **Macro-Averaged Calibration Scaling Across Training Budgets (M3-LLaVA 7B).** Evaluated across all 14 benchmarks over 5 subsampling seeds. **Bold:** best; *italic:* second best. ↓/↑: lower/higher is better.

Budget (N)	Calibration Method	Macro ECE (%) ↓	Macro Ada-ECE (%) ↓	Macro AUROC ↑	Macro Brier ↓
$N = 50$	Naive Confidence (NC)	40.33	40.33	0.671	0.3950
	Temperature Scaling (TS)	22.13	22.88	0.671	0.2219
	Platt Scaling (1D)	5.03	6.91	0.671	0.1392
	Trajectory Platt (5D)	<i>6.35</i>	<i>7.11</i>	<i>0.656</i>	<i>0.1416</i>
$N = 100$	Naive Confidence (NC)	40.33	40.33	0.671	0.3950
	Temperature Scaling (TS)	21.82	22.72	0.671	0.2215
	Platt Scaling (1D)	4.47	6.36	0.680	0.1373
	Trajectory Platt (5D)	<i>5.38</i>	<i>6.56</i>	<i>0.679</i>	<i>0.1379</i>
$N = 200$	Naive Confidence (NC)	40.33	40.33	0.671	0.3950
	Temperature Scaling (TS)	21.86	22.64	0.671	0.2212
	Platt Scaling (1D)	4.11	<i>5.94</i>	<i>0.683</i>	<i>0.1362</i>
	Trajectory Platt (5D)	<i>4.59</i>	5.92	0.695	0.1349
$N = 500$	Naive Confidence (NC)	40.33	40.33	0.671	0.3950
	Temperature Scaling (TS)	21.91	22.65	0.671	0.2213
	Platt Scaling (1D)	3.63	<i>5.86</i>	<i>0.682</i>	<i>0.1360</i>
	Trajectory Platt (5D)	<i>4.41</i>	5.57	0.704	0.1337
$N = 1000$	Naive Confidence (NC)	40.33	40.33	0.671	0.3950
	Temperature Scaling (TS)	21.92	22.68	0.671	0.2213
	Platt Scaling (1D)	3.36	<i>5.75</i>	<i>0.682</i>	<i>0.1358</i>
	Trajectory Platt (5D)	<i>4.12</i>	5.52	0.715	0.1327
$N = 1500$	Naive Confidence (NC)	40.33	40.33	0.671	0.3950
	Temperature Scaling (TS)	21.92	22.68	0.671	0.2213
	Platt Scaling (1D)	3.24	<i>5.71</i>	<i>0.682</i>	<i>0.1357</i>
	Trajectory Platt (5D)	<i>4.24</i>	5.52	0.715	0.1324
Full	Naive Confidence (NC)	40.33	40.33	0.671	0.3950
	Temperature Scaling (TS)	21.93	22.68	0.671	0.2213
	Platt Scaling (1D)	3.24	<i>5.70</i>	<i>0.682</i>	<i>0.1357</i>
	Trajectory Platt (5D)	<i>4.25</i>	5.49	0.717	0.1322

1.5 Group 7: Sample Efficiency & Low-Data Calibration Scaling

Table 7: **Macro-Averaged Calibration Scaling Across Training Budgets (MQT-LLaVA 7B).** Evaluated across all 14 benchmarks over 5 subsampling seeds. **Bold:** best; *italic:* second best. ↓/↑: lower/higher is better.

Budget (N)	Calibration Method	Macro ECE (%) ↓	Macro Ada-ECE (%) ↓	Macro AUROC ↑	Macro Brier ↓
$N = 50$	Naive Confidence (NC)	32.32	31.76	0.682	0.3055
	Temperature Scaling (TS)	23.88	24.36	0.682	0.2175
	Platt Scaling (1D)	6.83	<i>8.71</i>	<i>0.674</i>	0.1394
	Trajectory Platt (5D)	<i>7.49</i>	8.67	0.660	<i>0.1433</i>
$N = 100$	Naive Confidence (NC)	32.32	31.76	0.682	0.3055
	Temperature Scaling (TS)	23.73	24.24	0.682	0.2169
	Platt Scaling (1D)	6.61	<i>8.22</i>	<i>0.689</i>	0.1380
	Trajectory Platt (5D)	<i>6.70</i>	7.60	0.709	<i>0.1390</i>
$N = 200$	Naive Confidence (NC)	32.32	31.76	0.682	0.3055
	Temperature Scaling (TS)	23.53	24.21	0.682	0.2166
	Platt Scaling (1D)	5.84	<i>7.91</i>	<i>0.690</i>	<i>0.1369</i>
	Trajectory Platt (5D)	<i>6.02</i>	7.20	0.716	0.1360
$N = 500$	Naive Confidence (NC)	32.32	31.76	0.682	0.3055
	Temperature Scaling (TS)	23.44	24.15	0.682	0.2164
	Platt Scaling (1D)	5.46	<i>7.82</i>	<i>0.691</i>	<i>0.1366</i>
	Trajectory Platt (5D)	<i>5.80</i>	6.80	0.734	0.1339
$N = 1000$	Naive Confidence (NC)	32.32	31.76	0.682	0.3055
	Temperature Scaling (TS)	23.44	24.13	0.682	0.2163
	Platt Scaling (1D)	5.86	<i>7.80</i>	<i>0.686</i>	<i>0.1364</i>
	Trajectory Platt (5D)	<i>6.25</i>	7.15	0.748	0.1334
$N = 1500$	Naive Confidence (NC)	32.32	31.76	0.682	0.3055
	Temperature Scaling (TS)	23.44	24.12	0.682	0.2163
	Platt Scaling (1D)	5.83	<i>7.82</i>	<i>0.686</i>	<i>0.1364</i>
	Trajectory Platt (5D)	<i>6.12</i>	7.06	0.748	0.1334
Full	Naive Confidence (NC)	32.32	31.76	0.682	0.3055
	Temperature Scaling (TS)	23.45	24.13	0.682	0.2163
	Platt Scaling (1D)	5.88	<i>7.82</i>	<i>0.686</i>	<i>0.1365</i>
	Trajectory Platt (5D)	<i>6.06</i>	7.18	0.748	0.1334

Table 8: **Leave-One-Dataset-Out (LODO) Cross-Domain Transfer (M3-LLaVA 7B)**. Macro-averaged across all 14 held-out target benchmarks. Trained on pooled 13 benchmarks. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Transfer Protocol	Macro ECE (%) ↓	Macro Ada-ECE (%) ↓	Macro AUROC ↑	Macro Brier ↓
Naive Confidence (NC)	Zero-Shot Base	40.26	40.26	0.677	0.3976
Naive Confidence (NC)	Target Adapted (Saerens-EM)	51.89	51.89	0.599	0.5159
Temperature Scaling (TS)	Zero-Shot Base	<i>26.38</i>	<i>28.82</i>	0.677	<i>0.2491</i>
Temperature Scaling (TS)	Target Adapted (Saerens-EM)	52.62	52.62	0.677	0.5211
Platt Scaling (1D)	Zero-Shot Base	28.34	29.41	0.677	0.2536
Platt Scaling (1D)	Target Adapted (Saerens-EM)	40.88	40.88	0.677	0.4088
Trajectory Platt (5D)	Zero-Shot Base	24.55	24.59	<i>0.672</i>	0.2256
Trajectory Platt (5D)	Target Adapted (Saerens-EM)	29.58	29.59	<i>0.672</i>	0.2835

Table 9: **Leave-One-Dataset-Out (LODO) Cross-Domain Transfer (MQT-LLaVA 7B)**. Macro-averaged across all 14 held-out target benchmarks. Trained on pooled 13 benchmarks. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Transfer Protocol	Macro ECE (%) ↓	Macro Ada-ECE (%) ↓	Macro AUROC ↑	Macro Brier ↓
Naive Confidence (NC)	Zero-Shot Base	31.55	31.54	<i>0.698</i>	0.3070
Naive Confidence (NC)	Target Adapted (Saerens-EM)	53.34	53.34	0.629	0.5075
Temperature Scaling (TS)	Zero-Shot Base	28.15	28.44	<i>0.698</i>	0.2454
Temperature Scaling (TS)	Target Adapted (Saerens-EM)	58.44	58.44	<i>0.698</i>	0.5844
Platt Scaling (1D)	Zero-Shot Base	22.64	22.76	<i>0.698</i>	0.2147
Platt Scaling (1D)	Target Adapted (Saerens-EM)	26.15	26.16	<i>0.698</i>	0.2597
Trajectory Platt (5D)	Zero-Shot Base	<i>19.27</i>	<i>19.15</i>	0.699	0.1950
Trajectory Platt (5D)	Target Adapted (Saerens-EM)	19.12	19.04	0.699	<i>0.1992</i>

1.6 Group 4: Leave-One-Dataset-Out (LODO) Cross-Domain Transfer

2 Group 3: Temperature Transfer Robustness Ablation

Evaluating calibrators trained at $= 0.0$ and tested across $\in \{0.0, 0.3, 0.6, 1.0, 1.5\}$ and temperature Mean.

2.1 Temperature Transfer: Macro-Averaged Across 4 Benchmarks

Table 10: **Temperature Transfer Robustness (Macro-Averaged Across 4 Benchmarks) (M3-LLaVA 7B)**. Evaluated across sampling temperatures $T \in \{0.0, 0.3, 0.6, 1.0, 1.5\}$ and Mean (trained at $T = 0.0$). **Bold**: best; *italic*: second best. $\downarrow/\uparrow$: lower/higher is better.

Calibration Method	ECE (%) $\downarrow$	Ada-ECE (%) $\downarrow$	AUROC $\uparrow$	Brier $\downarrow$
Sampling Temperature $T = 0.0$				
Naive Confidence (NC)	10.03	10.18	<i>0.780</i>	0.1532
Temperature Scaling (TS)	9.16	9.74	<i>0.780</i>	0.1524
Platt Scaling (1D)	4.77	<i>6.40</i>	<i>0.780</i>	<i>0.1411</i>
Trajectory Platt (5D)	<i>4.80</i>	5.65	0.785	0.1402
Sampling Temperature $T = 0.3$				
Naive Confidence (NC)	23.98	23.28	<i>0.763</i>	0.2338
Temperature Scaling (TS)	22.14	21.60	<i>0.763</i>	0.2198
Platt Scaling (1D)	<i>19.62</i>	<i>19.05</i>	<i>0.763</i>	<i>0.2043</i>
Trajectory Platt (5D)	18.84	18.54	0.766	0.2013
Sampling Temperature $T = 0.6$				
Naive Confidence (NC)	16.05	15.37	<i>0.792</i>	0.1744
Temperature Scaling (TS)	13.87	13.62	<i>0.792</i>	0.1659
Platt Scaling (1D)	<i>9.65</i>	<i>9.66</i>	<i>0.792</i>	<i>0.1499</i>
Trajectory Platt (5D)	9.17	9.31	0.793	0.1484
Sampling Temperature $T = 1.0$				
Naive Confidence (NC)	9.82	9.86	0.832	0.1392
Temperature Scaling (TS)	9.41	9.34	0.832	0.1361
Platt Scaling (1D)	4.75	4.80	0.832	0.1238
Trajectory Platt (5D)	<i>5.92</i>	<i>5.43</i>	0.832	<i>0.1239</i>
Sampling Temperature $T = 1.5$				
Naive Confidence (NC)	<i>9.73</i>	<i>9.37</i>	<i>0.849</i>	0.1328
Temperature Scaling (TS)	11.33	10.78	<i>0.849</i>	0.1306
Platt Scaling (1D)	8.36	8.11	<i>0.849</i>	0.1244
Trajectory Platt (5D)	9.77	10.54	0.853	<i>0.1297</i>
Mean (Averaged Across Temperatures)				
Naive Confidence (NC)	13.92	13.61	<i>0.803</i>	0.1667
Temperature Scaling (TS)	13.18	13.02	<i>0.803</i>	<i>0.1610</i>
Platt Scaling (1D)	9.43	9.60	<i>0.803</i>	0.1487
Trajectory Platt (5D)	<i>9.70</i>	<i>9.90</i>	0.806	0.1487

Table 11: **Temperature Transfer Robustness (Macro-Averaged Across 4 Benchmarks) (MQT-LLaVA 7B)**. Evaluated across sampling temperatures $T \in \{0.0, 0.3, 0.6, 1.0, 1.5\}$ and Mean (trained at $T = 0.0$). **Bold**: best; *italic*: second best. $\downarrow/\uparrow$: lower/higher is better.

Calibration Method	ECE (%) $\downarrow$	Ada-ECE (%) $\downarrow$	AUROC $\uparrow$	Brier $\downarrow$
Sampling Temperature $T = 0.0$				
Naive Confidence (NC)	22.96	22.61	<i>0.769</i>	0.2304
Temperature Scaling (TS)	15.68	15.93	<i>0.769</i>	0.1944
Platt Scaling (1D)	<i>7.15</i>	<i>7.29</i>	<i>0.769</i>	<i>0.1625</i>
Trajectory Platt (5D)	4.90	6.22	0.779	0.1561
Sampling Temperature $T = 0.3$				
Naive Confidence (NC)	40.62	40.55	<i>0.744</i>	0.3834
Temperature Scaling (TS)	28.00	27.98	<i>0.744</i>	0.2677
Platt Scaling (1D)	<i>24.62</i>	<i>24.57</i>	<i>0.744</i>	<i>0.2517</i>
Trajectory Platt (5D)	20.65	20.66	0.773	0.2205
Sampling Temperature $T = 0.6$				
Naive Confidence (NC)	34.40	34.10	<i>0.762</i>	0.3146
Temperature Scaling (TS)	21.90	22.18	<i>0.762</i>	0.2256
Platt Scaling (1D)	<i>13.47</i>	<i>13.77</i>	<i>0.762</i>	<i>0.1876</i>
Trajectory Platt (5D)	10.41	10.76	0.791	0.1660
Sampling Temperature $T = 1.0$				
Naive Confidence (NC)	27.43	27.37	<i>0.780</i>	0.2547
Temperature Scaling (TS)	19.27	19.58	<i>0.780</i>	0.2061
Platt Scaling (1D)	<i>7.46</i>	<i>7.53</i>	<i>0.780</i>	<i>0.1585</i>
Trajectory Platt (5D)	5.00	5.09	0.810	0.1432
Sampling Temperature $T = 1.5$				
Naive Confidence (NC)	20.96	21.58	<i>0.785</i>	0.2100
Temperature Scaling (TS)	20.85	21.64	<i>0.785</i>	0.2001
Platt Scaling (1D)	<i>8.74</i>	<i>9.67</i>	<i>0.785</i>	<i>0.1514</i>
Trajectory Platt (5D)	7.23	8.48	0.796	0.1454
Mean (Averaged Across Temperatures)				
Naive Confidence (NC)	29.27	29.24	<i>0.768</i>	0.2786
Temperature Scaling (TS)	21.14	21.46	<i>0.768</i>	0.2188
Platt Scaling (1D)	<i>12.29</i>	<i>12.57</i>	<i>0.768</i>	<i>0.1823</i>
Trajectory Platt (5D)	9.64	10.24	0.790	0.1663

2.2 Temperature Transfer: POPE

Table 12: **Temperature Transfer Robustness on pope (M3-LLaVA 7B)**. Evaluated across sampling temperatures $T \in \{0.0, 0.3, 0.6, 1.0, 1.5\}$ and Mean (trained at $T = 0.0$). **Bold**: best; *italic*: second best. $\downarrow/\uparrow$: lower/higher is better.

Calibration Method	ECE (%) $\downarrow$	Ada-ECE (%) $\downarrow$	AUROC $\uparrow$	Brier $\downarrow$
Sampling Temperature $T = 0.0$				
Naive Confidence (NC)	4.03	<i>4.30</i>	<i>0.752</i>	<i>0.0915</i>
Temperature Scaling (TS)	3.67	4.23	<i>0.752</i>	0.0914
Platt Scaling (1D)	<i>3.69</i>	4.33	<i>0.752</i>	<i>0.0915</i>
Trajectory Platt (5D)	4.86	5.98	0.761	0.0921
Sampling Temperature $T = 0.3$				
Naive Confidence (NC)	<i>9.17</i>	8.63	<i>0.780</i>	<i>0.1001</i>
Temperature Scaling (TS)	9.01	<i>8.47</i>	<i>0.780</i>	0.0994
Platt Scaling (1D)	9.01	8.51	<i>0.780</i>	0.0994
Trajectory Platt (5D)	9.44	8.33	0.783	0.1004
Sampling Temperature $T = 0.6$				
Naive Confidence (NC)	7.17	6.05	<i>0.798</i>	<i>0.0948</i>
Temperature Scaling (TS)	7.03	<i>5.77</i>	<i>0.798</i>	0.0939
Platt Scaling (1D)	6.75	5.75	<i>0.798</i>	0.0939
Trajectory Platt (5D)	<i>7.01</i>	6.02	0.800	0.0949
Sampling Temperature $T = 1.0$				
Naive Confidence (NC)	4.81	<i>4.63</i>	<i>0.839</i>	0.0915
Temperature Scaling (TS)	5.25	4.90	<i>0.839</i>	0.0914
Platt Scaling (1D)	<i>5.02</i>	4.86	<i>0.839</i>	<i>0.0913</i>
Trajectory Platt (5D)	6.35	3.19	0.845	0.0904
Sampling Temperature $T = 1.5$				
Naive Confidence (NC)	5.68	5.35	<i>0.884</i>	0.0968
Temperature Scaling (TS)	<i>6.89</i>	<i>5.89</i>	<i>0.884</i>	0.0986
Platt Scaling (1D)	6.93	5.92	<i>0.884</i>	<i>0.0985</i>
Trajectory Platt (5D)	7.48	8.32	0.892	<i>0.0985</i>
Mean (Averaged Across Temperatures)				
Naive Confidence (NC)	6.17	5.79	<i>0.811</i>	0.0949
Temperature Scaling (TS)	6.37	<i>5.85</i>	<i>0.811</i>	0.0949
Platt Scaling (1D)	<i>6.28</i>	5.87	<i>0.811</i>	0.0949
Trajectory Platt (5D)	7.03	6.37	0.816	<i>0.0953</i>

Table 13: **Temperature Transfer Robustness on pope (MQT-LLaVA 7B)**. Evaluated across sampling temperatures $T \in \{0.0, 0.3, 0.6, 1.0, 1.5\}$ and Mean (trained at $T = 0.0$). **Bold**: best; *italic*: second best. $\downarrow/\uparrow$: lower/higher is better.

Calibration Method	ECE (%) $\downarrow$	Ada-ECE (%) $\downarrow$	AUROC $\uparrow$	Brier $\downarrow$
Sampling Temperature $T = 0.0$				
Naive Confidence (NC)	7.04	6.18	0.818	0.1030
Temperature Scaling (TS)	5.86	<i>5.44</i>	0.818	0.1014
Platt Scaling (1D)	<i>4.37</i>	4.79	0.818	0.1007
Trajectory Platt (5D)	3.95	5.59	0.818	<i>0.1008</i>
Sampling Temperature $T = 0.3$				
Naive Confidence (NC)	11.84	11.84	0.816	0.1246
Temperature Scaling (TS)	<i>10.86</i>	<i>10.78</i>	0.816	<i>0.1191</i>
Platt Scaling (1D)	10.66	10.54	0.816	0.1183
Trajectory Platt (5D)	11.35	11.38	0.816	0.1212
Sampling Temperature $T = 0.6$				
Naive Confidence (NC)	10.98	10.76	<i>0.813</i>	0.1283
Temperature Scaling (TS)	8.91	<i>9.00</i>	<i>0.813</i>	<i>0.1204</i>
Platt Scaling (1D)	<i>9.05</i>	8.99	<i>0.813</i>	0.1191
Trajectory Platt (5D)	9.24	10.07	0.814	0.1207
Sampling Temperature $T = 1.0$				
Naive Confidence (NC)	9.84	9.60	<i>0.804</i>	0.1397
Temperature Scaling (TS)	6.35	7.27	<i>0.804</i>	0.1323
Platt Scaling (1D)	<i>5.42</i>	<i>6.35</i>	<i>0.804</i>	0.1304
Trajectory Platt (5D)	5.19	5.63	0.805	<i>0.1306</i>
Sampling Temperature $T = 1.5$				
Naive Confidence (NC)	8.85	11.22	0.797	0.1590
Temperature Scaling (TS)	9.27	10.31	0.797	0.1530
Platt Scaling (1D)	6.32	<i>9.30</i>	0.797	0.1501
Trajectory Platt (5D)	<i>6.35</i>	8.84	<i>0.793</i>	<i>0.1509</i>
Mean (Averaged Across Temperatures)				
Naive Confidence (NC)	9.71	9.92	0.810	0.1309
Temperature Scaling (TS)	8.25	8.56	0.810	0.1252
Platt Scaling (1D)	7.17	8.00	0.810	0.1237
Trajectory Platt (5D)	<i>7.22</i>	<i>8.30</i>	<i>0.809</i>	<i>0.1248</i>

2.3 Temperature Transfer: ScienceQA

Table 14: **Temperature Transfer Robustness on scienceqa (M3-LLaVA 7B)**. Evaluated across sampling temperatures $T \in \{0.0, 0.3, 0.6, 1.0, 1.5\}$ and Mean (trained at $T = 0.0$). **Bold**: best; *italic*: second best. $\downarrow/\uparrow$: lower/higher is better.

Calibration Method	ECE (%) $\downarrow$	Ada-ECE (%) $\downarrow$	AUROC $\uparrow$	Brier $\downarrow$
Sampling Temperature $T = 0.0$				
Naive Confidence (NC)	11.08	10.66	<i>0.701</i>	0.1566
Temperature Scaling (TS)	8.69	9.76	<i>0.701</i>	0.1529
Platt Scaling (1D)	3.63	<i>7.46</i>	<i>0.701</i>	<i>0.1450</i>
Trajectory Platt (5D)	<i>3.73</i>	3.97	0.704	0.1446
Sampling Temperature $T = 0.3$				
Naive Confidence (NC)	18.42	17.16	<i>0.669</i>	0.1854
Temperature Scaling (TS)	14.86	13.94	<i>0.669</i>	0.1704
Platt Scaling (1D)	<i>12.97</i>	11.51	<i>0.669</i>	0.1577
Trajectory Platt (5D)	12.17	<i>12.55</i>	0.677	<i>0.1594</i>
Sampling Temperature $T = 0.6$				
Naive Confidence (NC)	16.66	16.40	<i>0.696</i>	0.1837
Temperature Scaling (TS)	11.47	11.57	<i>0.696</i>	0.1655
Platt Scaling (1D)	7.71	8.13	<i>0.696</i>	0.1496
Trajectory Platt (5D)	<i>8.15</i>	<i>9.21</i>	0.698	<i>0.1532</i>
Sampling Temperature $T = 1.0$				
Naive Confidence (NC)	13.94	13.60	0.729	0.1820
Temperature Scaling (TS)	11.42	11.14	0.729	0.1668
Platt Scaling (1D)	4.51	4.95	0.729	0.1482
Trajectory Platt (5D)	<i>6.06</i>	<i>6.75</i>	<i>0.723</i>	<i>0.1514</i>
Sampling Temperature $T = 1.5$				
Naive Confidence (NC)	18.68	18.38	<i>0.733</i>	0.2169
Temperature Scaling (TS)	19.71	18.94	<i>0.733</i>	0.1974
Platt Scaling (1D)	8.22	8.75	<i>0.733</i>	0.1609
Trajectory Platt (5D)	<i>10.45</i>	<i>11.45</i>	0.736	<i>0.1647</i>
Mean (Averaged Across Temperatures)				
Naive Confidence (NC)	15.76	15.24	<i>0.706</i>	0.1849
Temperature Scaling (TS)	13.23	13.07	<i>0.706</i>	0.1706
Platt Scaling (1D)	7.41	8.16	<i>0.706</i>	0.1523
Trajectory Platt (5D)	<i>8.11</i>	<i>8.78</i>	0.708	<i>0.1547</i>

Table 15: **Temperature Transfer Robustness on scienceqa (MQT-LLaVA 7B)**. Evaluated across sampling temperatures $T \in \{0.0, 0.3, 0.6, 1.0, 1.5\}$ and Mean (trained at $T = 0.0$). **Bold**: best; *italic*: second best. $\downarrow/\uparrow$: lower/higher is better.

Calibration Method	ECE (%) $\downarrow$	Ada-ECE (%) $\downarrow$	AUROC $\uparrow$	Brier $\downarrow$
Sampling Temperature $T = 0.0$				
Naive Confidence (NC)	23.29	23.04	<i>0.775</i>	0.2529
Temperature Scaling (TS)	10.47	10.84	<i>0.775</i>	0.1937
Platt Scaling (1D)	<i>7.75</i>	<i>6.78</i>	<i>0.775</i>	<i>0.1851</i>
Trajectory Platt (5D)	4.37	5.46	0.782	0.1798
Sampling Temperature $T = 0.3$				
Naive Confidence (NC)	33.59	33.29	<i>0.750</i>	0.3337
Temperature Scaling (TS)	22.74	22.74	<i>0.750</i>	<i>0.2522</i>
Platt Scaling (1D)	22.58	22.48	<i>0.750</i>	0.2505
Trajectory Platt (5D)	<i>22.66</i>	<i>22.65</i>	0.763	0.2526
Sampling Temperature $T = 0.6$				
Naive Confidence (NC)	29.71	28.74	<i>0.766</i>	0.2988
Temperature Scaling (TS)	14.42	15.14	<i>0.766</i>	0.2123
Platt Scaling (1D)	<i>12.54</i>	<i>12.74</i>	<i>0.766</i>	<i>0.2066</i>
Trajectory Platt (5D)	12.24	12.07	0.777	0.2023
Sampling Temperature $T = 1.0$				
Naive Confidence (NC)	24.28	24.28	<i>0.777</i>	0.2616
Temperature Scaling (TS)	11.71	12.08	<i>0.777</i>	0.1975
Platt Scaling (1D)	<i>7.50</i>	<i>7.67</i>	<i>0.777</i>	<i>0.1861</i>
Trajectory Platt (5D)	4.94	4.65	0.788	0.1796
Sampling Temperature $T = 1.5$				
Naive Confidence (NC)	17.13	17.24	<i>0.773</i>	0.2260
Temperature Scaling (TS)	11.59	13.85	<i>0.773</i>	0.1993
Platt Scaling (1D)	<i>12.70</i>	<i>13.51</i>	<i>0.773</i>	0.2025
Trajectory Platt (5D)	12.98	12.88	0.777	<i>0.2006</i>
Mean (Averaged Across Temperatures)				
Naive Confidence (NC)	25.60	25.32	<i>0.768</i>	0.2746
Temperature Scaling (TS)	14.18	14.93	<i>0.768</i>	0.2110
Platt Scaling (1D)	<i>12.61</i>	<i>12.64</i>	<i>0.768</i>	<i>0.2062</i>
Trajectory Platt (5D)	11.44	11.54	0.777	0.2030

2.4 Temperature Transfer: TextVQA

Table 16: **Temperature Transfer Robustness on textvqa (M3-LLaVA 7B)**. Evaluated across sampling temperatures $T \in \{0.0, 0.3, 0.6, 1.0, 1.5\}$ and Mean (trained at $T = 0.0$). **Bold**: best; *italic*: second best. $\downarrow/\uparrow$: lower/higher is better.

Calibration Method	ECE (%) $\downarrow$	Ada-ECE (%) $\downarrow$	AUROC $\uparrow$	Brier $\downarrow$
Sampling Temperature $T = 0.0$				
Naive Confidence (NC)	8.32	9.07	<i>0.832</i>	0.1715
Temperature Scaling (TS)	8.70	8.92	<i>0.832</i>	0.1709
Platt Scaling (1D)	<i>5.01</i>	<i>6.11</i>	<i>0.832</i>	<i>0.1637</i>
Trajectory Platt (5D)	4.35	5.91	0.839	0.1604
Sampling Temperature $T = 0.3$				
Naive Confidence (NC)	25.60	24.89	<i>0.810</i>	0.2595
Temperature Scaling (TS)	24.77	24.35	<i>0.810</i>	0.2545
Platt Scaling (1D)	<i>22.86</i>	<i>22.69</i>	<i>0.810</i>	<i>0.2446</i>
Trajectory Platt (5D)	20.49	20.19	0.815	0.2292
Sampling Temperature $T = 0.6$				
Naive Confidence (NC)	15.71	15.40	<i>0.843</i>	0.1901
Temperature Scaling (TS)	14.74	14.90	<i>0.843</i>	0.1875
Platt Scaling (1D)	<i>12.64</i>	<i>12.78</i>	<i>0.843</i>	<i>0.1761</i>
Trajectory Platt (5D)	9.30	8.79	0.848	0.1646
Sampling Temperature $T = 1.0$				
Naive Confidence (NC)	7.40	7.86	<i>0.894</i>	0.1340
Temperature Scaling (TS)	7.62	7.89	<i>0.894</i>	0.1341
Platt Scaling (1D)	4.14	4.12	<i>0.894</i>	<i>0.1269</i>
Trajectory Platt (5D)	<i>5.81</i>	<i>6.13</i>	0.896	0.1254
Sampling Temperature $T = 1.5$				
Naive Confidence (NC)	7.83	7.59	<i>0.886</i>	0.1142
Temperature Scaling (TS)	<i>8.07</i>	<i>7.97</i>	<i>0.886</i>	<i>0.1152</i>
Platt Scaling (1D)	9.70	9.54	<i>0.886</i>	0.1222
Trajectory Platt (5D)	13.60	13.38	0.892	0.1392
Mean (Averaged Across Temperatures)				
Naive Confidence (NC)	12.97	12.96	<i>0.853</i>	0.1739
Temperature Scaling (TS)	12.78	12.81	<i>0.853</i>	0.1724
Platt Scaling (1D)	<i>10.87</i>	<i>11.05</i>	<i>0.853</i>	<i>0.1667</i>
Trajectory Platt (5D)	10.71	10.88	0.858	0.1637

Table 17: **Temperature Transfer Robustness on textvqa (MQT-LLaVA 7B)**. Evaluated across sampling temperatures $T \in \{0.0, 0.3, 0.6, 1.0, 1.5\}$ and Mean (trained at $T = 0.0$). **Bold**: best; *italic*: second best. $\downarrow/\uparrow$: lower/higher is better.

Calibration Method	ECE (%) $\downarrow$	Ada-ECE (%) $\downarrow$	AUROC $\uparrow$	Brier $\downarrow$
Sampling Temperature $T = 0.0$				
Naive Confidence (NC)	31.24	30.96	<i>0.681</i>	0.3237
Temperature Scaling (TS)	17.62	18.67	<i>0.681</i>	0.2497
Platt Scaling (1D)	<i>9.78</i>	<i>11.82</i>	<i>0.681</i>	<i>0.2164</i>
Trajectory Platt (5D)	5.87	7.83	0.714	0.1985
Sampling Temperature $T = 0.3$				
Naive Confidence (NC)	54.81	54.81	<i>0.651</i>	0.5199
Temperature Scaling (TS)	28.32	28.32	<i>0.651</i>	<i>0.2957</i>
Platt Scaling (1D)	<i>28.11</i>	<i>28.11</i>	<i>0.651</i>	0.3156
Trajectory Platt (5D)	16.48	16.54	0.750	0.2155
Sampling Temperature $T = 0.6$				
Naive Confidence (NC)	47.84	47.84	<i>0.673</i>	0.4377
Temperature Scaling (TS)	24.97	25.27	<i>0.673</i>	0.2715
Platt Scaling (1D)	<i>15.58</i>	<i>16.73</i>	<i>0.673</i>	<i>0.2451</i>
Trajectory Platt (5D)	8.37	8.49	0.769	0.1790
Sampling Temperature $T = 1.0$				
Naive Confidence (NC)	40.67	40.67	<i>0.720</i>	0.3603
Temperature Scaling (TS)	26.22	26.22	<i>0.720</i>	0.2546
Platt Scaling (1D)	<i>9.87</i>	<i>9.01</i>	<i>0.720</i>	<i>0.1855</i>
Trajectory Platt (5D)	5.45	4.79	0.813	0.1397
Sampling Temperature $T = 1.5$				
Naive Confidence (NC)	33.47	33.47	<i>0.707</i>	0.2937
Temperature Scaling (TS)	31.34	31.18	<i>0.707</i>	0.2472
Platt Scaling (1D)	<i>8.30</i>	<i>9.18</i>	<i>0.707</i>	<i>0.1518</i>
Trajectory Platt (5D)	2.92	4.58	0.745	0.1291
Mean (Averaged Across Temperatures)				
Naive Confidence (NC)	41.61	41.55	<i>0.686</i>	0.3871
Temperature Scaling (TS)	25.69	25.93	<i>0.686</i>	0.2638
Platt Scaling (1D)	<i>14.33</i>	<i>14.97</i>	<i>0.686</i>	<i>0.2229</i>
Trajectory Platt (5D)	7.82	8.45	0.758	0.1724

2.5 Temperature Transfer: VizWiz-VQA

Table 18: **Temperature Transfer Robustness on vizwiz-vqa (M3-LLaVA 7B)**. Evaluated across sampling temperatures $T \in \{0.0, 0.3, 0.6, 1.0, 1.5\}$ and Mean (trained at $T = 0.0$). **Bold**: best; *italic*: second best. $\downarrow/\uparrow$: lower/higher is better.

Calibration Method	ECE (%) $\downarrow$	Ada-ECE (%) $\downarrow$	AUROC $\uparrow$	Brier $\downarrow$
Sampling Temperature $T = 0.0$				
Naive Confidence (NC)	16.71	16.71	<i>0.835</i>	0.1933
Temperature Scaling (TS)	15.56	16.03	<i>0.835</i>	0.1945
Platt Scaling (1D)	<i>6.76</i>	<i>7.69</i>	<i>0.835</i>	<i>0.1642</i>
Trajectory Platt (5D)	6.27	6.75	0.837	0.1636
Sampling Temperature $T = 0.3$				
Naive Confidence (NC)	42.74	42.43	0.793	0.3901
Temperature Scaling (TS)	39.93	39.64	0.793	0.3551
Platt Scaling (1D)	<i>33.64</i>	<i>33.47</i>	0.793	0.3155
Trajectory Platt (5D)	33.27	33.10	<i>0.791</i>	<i>0.3163</i>
Sampling Temperature $T = 0.6$				
Naive Confidence (NC)	24.67	23.63	0.829	0.2288
Temperature Scaling (TS)	22.24	22.24	0.829	0.2168
Platt Scaling (1D)	11.49	11.99	0.829	0.1799
Trajectory Platt (5D)	<i>12.22</i>	<i>13.23</i>	<i>0.827</i>	<i>0.1810</i>
Sampling Temperature $T = 1.0$				
Naive Confidence (NC)	13.12	13.36	0.864	0.1493
Temperature Scaling (TS)	13.36	13.44	0.864	0.1522
Platt Scaling (1D)	5.31	5.28	0.864	<i>0.1289</i>
Trajectory Platt (5D)	<i>5.46</i>	<i>5.66</i>	<i>0.863</i>	0.1284
Sampling Temperature $T = 1.5$				
Naive Confidence (NC)	6.71	6.16	<i>0.892</i>	0.1031
Temperature Scaling (TS)	10.63	10.32	<i>0.892</i>	<i>0.1112</i>
Platt Scaling (1D)	8.60	<i>8.23</i>	<i>0.892</i>	0.1159
Trajectory Platt (5D)	<i>7.53</i>	9.01	0.893	0.1163
Mean (Averaged Across Temperatures)				
Naive Confidence (NC)	20.79	20.46	0.843	0.2129
Temperature Scaling (TS)	20.34	20.34	0.843	0.2060
Platt Scaling (1D)	<i>13.16</i>	13.33	0.843	0.1809
Trajectory Platt (5D)	12.95	<i>13.55</i>	<i>0.842</i>	<i>0.1811</i>

Table 19: **Temperature Transfer Robustness on vizwiz-vqa (MQT-LLaVA 7B)**. Evaluated across sampling temperatures $T \in \{0.0, 0.3, 0.6, 1.0, 1.5\}$ and Mean (trained at $T = 0.0$). **Bold**: best; *italic*: second best. $\downarrow/\uparrow$: lower/higher is better.

Calibration Method	ECE (%) $\downarrow$	Ada-ECE (%) $\downarrow$	AUROC $\uparrow$	Brier $\downarrow$
Sampling Temperature $T = 0.0$				
Naive Confidence (NC)	30.26	30.26	<i>0.802</i>	0.2421
Temperature Scaling (TS)	28.77	28.77	<i>0.802</i>	0.2326
Platt Scaling (1D)	<i>6.70</i>	5.76	<i>0.802</i>	<i>0.1479</i>
Trajectory Platt (5D)	5.41	<i>5.98</i>	0.803	0.1454
Sampling Temperature $T = 0.3$				
Naive Confidence (NC)	62.25	62.25	<i>0.761</i>	0.5552
Temperature Scaling (TS)	50.10	50.10	<i>0.761</i>	0.4037
Platt Scaling (1D)	<i>37.14</i>	<i>37.14</i>	<i>0.761</i>	<i>0.3223</i>
Trajectory Platt (5D)	32.11	32.08	0.763	0.2928
Sampling Temperature $T = 0.6$				
Naive Confidence (NC)	49.08	49.08	<i>0.795</i>	0.3935
Temperature Scaling (TS)	39.31	39.31	<i>0.795</i>	0.2980
Platt Scaling (1D)	<i>16.71</i>	<i>16.63</i>	<i>0.795</i>	<i>0.1798</i>
Trajectory Platt (5D)	11.79	12.42	0.805	0.1621
Sampling Temperature $T = 1.0$				
Naive Confidence (NC)	34.94	34.94	<i>0.820</i>	0.2574
Temperature Scaling (TS)	32.80	32.74	<i>0.820</i>	0.2402
Platt Scaling (1D)	<i>7.03</i>	<i>7.09</i>	<i>0.820</i>	<i>0.1318</i>
Trajectory Platt (5D)	4.44	5.28	0.832	0.1229
Sampling Temperature $T = 1.5$				
Naive Confidence (NC)	24.39	24.39	<i>0.864</i>	0.1613
Temperature Scaling (TS)	31.20	31.20	<i>0.864</i>	0.2008
Platt Scaling (1D)	<i>7.66</i>	6.67	<i>0.864</i>	0.1011
Trajectory Platt (5D)	6.69	<i>7.64</i>	0.869	<i>0.1012</i>
Mean (Averaged Across Temperatures)				
Naive Confidence (NC)	40.18	40.18	<i>0.808</i>	0.3219
Temperature Scaling (TS)	36.44	36.42	<i>0.808</i>	0.2751
Platt Scaling (1D)	<i>15.05</i>	<i>14.66</i>	<i>0.808</i>	<i>0.1766</i>
Trajectory Platt (5D)	12.09	12.68	0.814	0.1649

Table 20: **Calibration Benchmark on ai2d (M3-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	AUROC ↑	Brier ↓
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	41.80	41.80	0.598	0.4180
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	8.56	9.38	0.598	0.2474
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	3.49	5.19	0.598	<i>0.2394</i>
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	<i>7.21</i>	<i>8.83</i>	0.645	0.2307
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	29.40	-	<i>0.617</i>	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	31.60	-	0.604	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	27.20	-	0.607	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	28.60	-	0.608	-

Table 21: **Calibration Benchmark on ai2d (MQT-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	AUROC ↑	Brier ↓
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	24.34	24.32	0.705	0.2862
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	<i>11.04</i>	<i>13.01</i>	0.705	0.2282
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	7.89	9.51	0.705	0.2172
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	14.16	13.41	<i>0.698</i>	<i>0.2175</i>
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	32.40	-	0.611	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	35.70	-	0.594	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	31.50	-	0.639	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	31.80	-	0.612	-

3 Group 1: Per-Dataset In-Domain Calibration Benchmarks

Individual benchmark tables for all 14 vision-language datasets under the 8-method, 4-metric schema.

3.1 Benchmark: AI2D

Table 22: **Calibration Benchmark on chartqa (M3-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	AUROC ↑	Brier ↓
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	70.70	70.70	0.835	0.6077
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	45.45	45.07	0.835	0.2775
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	2.96	2.35	0.835	<i>0.0685</i>
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	<i>3.43</i>	<i>3.88</i>	0.826	0.0681
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	8.90	-	0.734	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	25.80	-	0.790	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	19.30	-	0.818	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	8.80	-	<i>0.828</i>	-

Table 23: **Calibration Benchmark on chartqa (MQT-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	AUROC ↑	Brier ↓
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	62.99	62.99	0.467	0.5705
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	41.54	41.54	0.467	0.2718
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	<i>2.83</i>	<i>8.03</i>	0.533	<i>0.0980</i>
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	2.36	6.27	0.713	0.0920
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	14.80	-	0.710	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	37.70	-	0.830	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	24.90	-	<i>0.855</i>	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	10.00	-	0.861	-

3.2 Benchmark: CHARTQA

Table 24: **Calibration Benchmark on docvqa (M3-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	AUROC ↑	Brier ↓
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	73.52	73.52	0.732	0.6320
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	48.86	<i>48.86</i>	0.732	0.2796
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	1.80	2.39	0.732	<i>0.0418</i>
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	<i>2.31</i>	2.39	0.732	0.0409
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	15.80	-	0.786	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	37.10	-	0.809	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	29.70	-	<i>0.830</i>	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	10.40	-	0.834	-

Table 25: **Calibration Benchmark on docvqa (MQT-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	AUROC ↑	Brier ↓
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	51.18	51.18	0.644	0.3845
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	46.63	46.63	0.644	0.2557
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	0.47	<i>4.56</i>	0.644	<i>0.0383</i>
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	<i>1.05</i>	3.86	0.585	0.0381
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	12.70	-	0.805	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	39.30	-	0.816	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	33.50	-	<i>0.839</i>	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	14.00	-	0.853	-

3.3 Benchmark: DOCVQA

Table 26: **Calibration Benchmark on gqa (M3-LLaVA 7B).** Bold: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	AUROC ↑	Brier ↓
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	32.54	32.43	0.685	0.3221
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	8.24	9.90	0.685	0.2076
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	4.08	<i>6.70</i>	0.685	<i>0.1988</i>
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	<i>6.59</i>	6.31	<i>0.706</i>	0.1965
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	18.80	-	0.696	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	25.30	-	0.623	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	17.60	-	0.690	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	17.40	-	0.712	-

Table 27: **Calibration Benchmark on gqa (MQT-LLaVA 7B).** Bold: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	AUROC ↑	Brier ↓
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	13.64	10.19	<i>0.757</i>	0.1865
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	9.21	10.61	<i>0.757</i>	<i>0.1860</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	<i>9.56</i>	<i>10.60</i>	<i>0.757</i>	0.1861
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	13.81	14.66	0.763	0.1854
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	19.80	-	0.698	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	30.00	-	0.624	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	18.70	-	0.696	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	16.70	-	0.715	-

3.4 Benchmark: GQA

Table 28: **Calibration Benchmark on infographicvqa (M3-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	AUROC ↑	Brier ↓
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	84.55	84.55	0.425	0.7641
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	53.91	53.91	0.425	<i>0.3000</i>
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	<i>0.04</i>	1.13	<i>0.575</i>	0.0079
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	0.03	<i>1.25</i>	0.649	0.0079
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	16.50	-	0.547	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	39.20	-	0.534	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	34.80	-	0.561	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	16.60	-	0.562	-

Table 29: **Calibration Benchmark on infographicvqa (MQT-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	AUROC ↑	Brier ↓
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	75.51	75.51	<i>0.769</i>	0.6535
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	52.63	52.63	<i>0.769</i>	0.2826
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	0.28	<i>1.11</i>	<i>0.769</i>	<i>0.0050</i>
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	<i>0.30</i>	1.08	0.930	0.0049
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	34.10	-	0.615	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	47.50	-	0.648	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	47.80	-	0.648	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	26.10	-	0.646	-

3.5 Benchmark: INFOGRAPHICVQA

Table 30: **Calibration Benchmark on lego-puzzles (M3-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	AUROC ↑	Brier ↓
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	73.16	73.16	0.477	0.7314
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	34.71	34.71	0.477	0.3160
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	0.71	<i>11.89</i>	0.477	<i>0.1958</i>
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	<i>4.15</i>	9.40	0.601	0.1914
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	21.20	-	0.551	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	25.00	-	0.582	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	15.00	-	0.571	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	11.00	-	<i>0.588</i>	-

Table 31: **Calibration Benchmark on lego-puzzles (MQT-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	AUROC ↑	Brier ↓
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	34.60	33.80	0.510	0.3474
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	23.68	23.68	0.510	0.2539
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	0.60	<i>12.43</i>	0.490	<i>0.1968</i>
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	<i>7.81</i>	8.67	0.593	0.1924
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	25.30	-	0.531	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	45.20	-	0.533	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	39.40	-	0.502	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	27.60	-	<i>0.537</i>	-

3.6 Benchmark: LEGOPUZZLES

Table 32: **Calibration Benchmark on mmbench (M3-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	AUROC ↑	Brier ↓
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	28.62	28.58	0.629	0.2860
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	6.68	7.88	0.629	0.2014
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	3.07	5.50	0.629	<i>0.1987</i>
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	<i>5.13</i>	<i>5.72</i>	0.691	0.1873
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	20.50	-	0.660	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	22.00	-	0.646	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	18.20	-	<i>0.681</i>	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	19.30	-	0.658	-

Table 33: **Calibration Benchmark on mmbench (MQT-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	AUROC ↑	Brier ↓
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	15.38	15.27	0.828	0.1866
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	<i>9.81</i>	10.24	0.828	0.1560
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	8.11	<i>8.38</i>	0.828	0.1503
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	10.42	7.51	<i>0.819</i>	<i>0.1543</i>
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	22.20	-	0.613	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	23.30	-	0.596	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	20.90	-	0.675	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	22.10	-	0.618	-

3.7 Benchmark: MMBENCH

Table 34: **Calibration Benchmark on mmmu (M3-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	AUROC ↑	Brier ↓
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	78.76	78.76	0.639	0.6889
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	52.21	52.21	0.639	0.2908
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	<i>1.81</i>	3.44	0.639	<i>0.0205</i>
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	1.68	<i>3.56</i>	<i>0.623</i>	0.0198
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	36.40	-	0.566	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	47.60	-	0.568	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	50.10	-	0.602	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	39.70	-	0.592	-

Table 35: **Calibration Benchmark on mmmu (MQT-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	AUROC ↑	Brier ↓
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	52.24	52.24	0.315	0.3669
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	49.32	49.32	0.315	0.2556
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	0.16	<i>2.14</i>	0.315	0.0115
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	<i>0.23</i>	2.04	0.724	<i>0.0116</i>
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	39.10	-	0.584	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	44.40	-	0.588	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	45.50	-	<i>0.627</i>	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	37.70	-	0.598	-

3.8 Benchmark: MMMU

Table 36: **Calibration Benchmark on pope (M3-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	AUROC ↑	Brier ↓
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	4.03	<i>4.30</i>	<i>0.752</i>	<i>0.0915</i>
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	3.67	4.23	<i>0.752</i>	0.0914
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	<i>3.69</i>	4.33	<i>0.752</i>	<i>0.0915</i>
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	4.86	5.98	0.761	0.0921
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	12.80	-	0.668	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	14.90	-	0.387	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	16.50	-	0.678	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	13.50	-	0.666	-

Table 37: **Calibration Benchmark on pope (MQT-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	AUROC ↑	Brier ↓
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	7.04	6.18	0.818	0.1030
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	5.86	<i>5.44</i>	0.818	0.1014
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	<i>4.37</i>	4.79	0.818	0.1007
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	3.95	5.59	0.818	<i>0.1008</i>
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	13.90	-	0.683	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	16.70	-	0.368	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	18.10	-	0.616	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	14.70	-	<i>0.693</i>	-

3.9 Benchmark: POPE

Table 38: **Calibration Benchmark on scienceqa (M3-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	AUROC ↑	Brier ↓
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	11.08	10.66	<i>0.701</i>	0.1566
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	8.69	9.76	<i>0.701</i>	0.1529
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	3.63	<i>7.46</i>	<i>0.701</i>	<i>0.1450</i>
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	<i>3.73</i>	3.97	0.704	0.1446
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	26.80	-	0.659	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	30.00	-	0.633	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	30.10	-	0.636	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	25.10	-	0.662	-

Table 39: **Calibration Benchmark on scienceqa (MQT-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	AUROC ↑	Brier ↓
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	23.29	23.04	<i>0.775</i>	0.2529
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	10.47	10.84	<i>0.775</i>	0.1937
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	<i>7.75</i>	<i>6.78</i>	<i>0.775</i>	<i>0.1851</i>
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	4.37	5.46	0.782	0.1798
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	30.80	-	0.581	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	32.50	-	0.568	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	29.00	-	0.588	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	30.20	-	0.581	-

3.10 Benchmark: SCIENCEQA

Table 40: **Calibration Benchmark on seedbench (M3-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	AUROC ↑	Brier ↓
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	32.40	32.40	0.500	0.3240
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	<i>1.31</i>	6.16	0.500	0.2189
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	1.17	<i>5.98</i>	0.500	<i>0.2188</i>
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	3.43	5.66	0.641	0.2062
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	23.30	-	<i>0.644</i>	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	25.00	-	0.629	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	22.70	-	0.655	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	22.40	-	<i>0.644</i>	-

Table 41: **Calibration Benchmark on seedbench (MQT-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	AUROC ↑	Brier ↓
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	18.56	17.74	<i>0.733</i>	0.2297
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	<i>6.68</i>	10.44	<i>0.733</i>	0.2015
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	9.51	8.62	<i>0.733</i>	<i>0.1986</i>
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	6.66	<i>9.67</i>	0.741	0.1957
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	28.10	-	0.610	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	30.00	-	0.599	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	27.00	-	0.639	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	27.70	-	0.605	-

3.11 Benchmark: SEEDBENCH

Table 42: **Calibration Benchmark on textvqa (M3-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	AUROC ↑	Brier ↓
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	8.32	9.07	<i>0.832</i>	0.1715
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	8.70	8.92	<i>0.832</i>	0.1709
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	<i>5.01</i>	<i>6.11</i>	<i>0.832</i>	<i>0.1637</i>
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	4.35	5.91	0.839	0.1604
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	11.80	-	0.811	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	28.20	-	0.774	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	26.60	-	0.814	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	10.40	-	0.829	-

Table 43: **Calibration Benchmark on textvqa (MQT-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	AUROC ↑	Brier ↓
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	31.24	30.96	0.681	0.3237
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	17.62	18.67	0.681	0.2497
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	<i>9.78</i>	<i>11.82</i>	0.681	<i>0.2164</i>
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	5.87	7.83	0.714	0.1985
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	14.80	-	0.793	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	31.50	-	0.789	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	28.30	-	<i>0.802</i>	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	12.10	-	0.818	-

3.12 Benchmark: TEXTVQA

Table 44: **Calibration Benchmark on vizwiz-vqa (M3-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	AUROC ↑	Brier ↓
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	16.71	16.71	<i>0.835</i>	0.1933
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	15.56	16.03	<i>0.835</i>	0.1945
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	<i>6.76</i>	<i>7.69</i>	<i>0.835</i>	<i>0.1642</i>
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	6.27	6.75	0.837	0.1636
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	14.80	-	0.723	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	32.40	-	0.701	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	32.30	-	0.757	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	13.70	-	0.768	-

Table 45: **Calibration Benchmark on vizwiz-vqa (MQT-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	AUROC ↑	Brier ↓
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	30.26	30.26	<i>0.802</i>	0.2421
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	28.77	28.77	<i>0.802</i>	0.2326
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	<i>6.70</i>	5.76	<i>0.802</i>	<i>0.1479</i>
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	5.41	<i>5.98</i>	0.803	0.1454
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	18.50	-	0.718	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	39.70	-	0.689	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	39.20	-	0.744	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	19.30	-	0.754	-

3.13 Benchmark: VIZWIZVQA

Table 46: **Calibration Benchmark on vqav2 (M3-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	AUROC ↑	Brier ↓
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	8.51	<i>8.06</i>	0.756	<i>0.1427</i>
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	10.48	10.51	0.756	0.1495
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	7.13	9.71	0.756	0.1455
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	6.30	7.19	0.787	0.1418
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	12.40	-	0.787	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	11.40	-	0.690	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	<i>6.20</i>	-	<i>0.816</i>	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	5.80	-	0.819	-

Table 47: **Calibration Benchmark on vqav2 (MQT-LLaVA 7B)**. **Bold**: best; *italic*: second best. ↓/↑: lower/higher is better.

Calibration Method	Regime / Sampling	ECE (%) ↓	Ada-ECE (%) ↓	AUROC ↑	Brier ↓
Naive Confidence (NC)	Single-Pass ($T = 0.0$)	12.14	<i>10.93</i>	0.746	0.1439
Temperature Scaling (TS)	Single-Pass ($T = 0.0$)	15.10	16.04	0.746	0.1597
Platt Scaling (1D)	Single-Pass ($T = 0.0$)	14.26	14.95	0.746	0.1588
Trajectory Platt (5D)	Single-Pass ($T = 0.0$)	<i>8.48</i>	8.52	0.791	<i>0.1514</i>
LN-Entropy	Multi-Pass ($T = 0.5, K = 10$)	12.30	-	0.767	-
Semantic Entropy	Multi-Pass ($T = 0.5, K = 10$)	13.70	-	0.699	-
EigenScore	Multi-Pass ($T = 0.5, K = 10$)	10.20	-	0.773	-
UMPIRE	Multi-Pass ($T = 0.5, K = 10$)	7.80	-	<i>0.790</i>	-

3.14 Benchmark: VQAV2